

Kaylee Yaxuan Li

Ph.D. Candidate | Computer Science and Engineering | University of Michigan
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Research Interests

My research develops wearable and ubiquitous AI systems that interpret human activity and context from multimodal data. I focus on sensing and efficient semantic representations of daily life, long-horizon memory architectures for scalable reasoning, and proactive assistance on resource-constrained wearable and mobile platforms. My work seeks to transform everyday human experience into persistent, actionable intelligence that augments human perception, memory, reasoning, and decision-making, with broader implications for physical AI and human-agent interaction.

Education

- 2023–Present **University of Michigan**, Ann Arbor, MI, USA
Ph.D. in Computer Science and Engineering
Advisor: Prof. Alanson P. Sample; Prof. Kang G. Shin
- 2019–2022 **McGill University**, Montreal, Quebec, Canada
M.Sc. in Electrical Engineering (Research-Based)
Advisor: Prof. Jeremy R. Cooperstock
- 2015–2019 **Shandong University**, Shandong, China
B.Eng. in Control Science and Engineering

Industry Experience

- 2025 **Meta Reality Labs**, Seattle, WA, USA
Research Scientist Intern Jun. 2025 – Dec. 2025
Developed keyframe extraction methods for long-form egocentric video on Meta AI glasses.
- 2021 **Samasource Impact Sourcing, Inc.**, Canada
Machine Learning Intern Jan. 2021 – May 2021
Developed an image and video annotation platform.

Honors & Fellowships

- 2025 **Rackham Doctoral Intern Fellowship**, University of Michigan
- 2024 **Rackham Chia-Lun Lo Fellowship**, University of Michigan
- 2024 **Qualcomm Fellowship**, Qualcomm Technologies, Inc.

Publications

- 2026 **GOHi: Goal-Oriented Highlight Extraction for Long-Form Egocentric Video**
Kaylee Yaxuan Li, Ke Sun, Kang G. Shin, Alanson P. Sample, Amy Karlson, Eric Whitmire, Yifan Wang. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (IMWUT/UbiComp).
- 2026 **EchoScriptor: Automatic Lifelogging Narratives via Activity-Based Audio-Language Model**
Kaylee Yaxuan Li, Xinghao Zhou, Haizhong Zheng, Kang G. Shin, Alanson P. Sample. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI)*.
- 2026 **AutoCLC: Towards Automated Assessment and Feedback on Closed-Loop Communication in Team-based Healthcare Simulation Training**
Kaylee Yaxuan Li^{*}, Kunpeng Huang^{*}, Bryan Harmer, Casey Ducharme, Benjamin Heasman, Eleanor Falahee, James M. Cooke, Michael Cole, Alanson P. Sample, Vitaliy Popov. *ACM Transactions on Computing for Healthcare*, 7(1), 1–31.

- 2025 **Leveraging AI for Automated Evaluation of Closed-Loop Communication in VR-based Acute Care Team Training**
Vitaliy Popov, **Kaylee Yaxuan Li**, Kunpeng Huang, Alanson P. Sample, Bryan Harmer, James Cooke, Michael Cole. *18th International Conference on Computer-Supported Collaborative Learning (CSCL)*.
- 2025 **HandSAW: Wearable Hand-based Event Recognition via On-Body Surface Acoustic Waves**
Kaylee Yaxuan Li, Yasha Iravantchi, Yichen Zhu, Hyunmin Park, Alanson P. Sample. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)*.
- 2024 **ECG Signal Construction From Heart Sounds via Single Node, Surface Acoustic Sensing**
Kaylee Yaxuan Li, Yasha Iravantchi, Hyunmin Park, Yiming Liu, Alanson P. Sample. *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.
- 2022 **BodyTrak: Inferring Full-body Poses from Body Silhouettes Using a Miniature Camera on a Wristband**
Hyunchul Lim, **Yaxuan Li**, Matthew Dressa, Fang Hu, Jae Hoon Kim, Ruidong Zhang, Cheng Zhang. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)*.
- 2022 **Deep Learning in Human Activity Recognition with Wearable Sensors: A Review on Advances**
Shibo Zhang*, **Yaxuan Li***, Shen Zhang, Farzad Shahabi, Stephen Xia, Yu Deng, Nabil Alshurafa. *Sensors*, 22(4), 1476. **Best Paper Award**.
- 2021 **SpeeChin: A Smart Necklace for Silent Speech Recognition**
Ruidong Zhang, Mingyang Chen, Benjamin Steeper, **Yaxuan Li**, Zihan Yan, Yizhuo Chen, Songyun Tao, Tuochao Chen, Hyunchul Lim, Cheng Zhang. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)*.
- 2021 **Towards Context-aware Automatic Haptic Effect Generation for Home Theatre Environments**
Yaxuan Li, Yongjae Yoo, Antoine Weill-Duflos, Jeremy R. Cooperstock. *27th ACM Symposium on Virtual Reality Software and Technology (VRST)*.
- 2021 **NeckFace: Continuously Tracking Full Facial Expressions on Neck-mounted Wearables**
Tuochao Chen*, **Yaxuan Li***, Songyun Tao*, Hyunchul Lim, Mose Sakashita, Ruidong Zhang, Francois Guimbretiere, Cheng Zhang. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)*.
- 2021 **Context-aware Automatic Haptic Effect Generation Algorithm for Improved Content Viewing Experiences**
Yaxuan Li, Yongjae Yoo, Antoine Weill-Duflos, Jeremy R. Cooperstock. *IEEE World Haptics Conference (WHC), Work-in-Progress*.
- 2019 **Safe Navigation of Quadrotors with Jerk Limited Trajectory**
Shupeng Lai, Menglu Lan, **Yaxuan Li**, Ben M. Chen. *Frontiers of Information Technology & Electronic Engineering*, 20(1), 107–119.

Research Experience

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| 2023–Present | University of Michigan
Graduate Student Research Assistant – Wearable and Ubiquitous AI
Advisor: Prof. Alanson P. Sample; Prof. Kang G. Shin.
Develop wearable AI systems spanning novel sensing methods, multimodal semantic representation, and long-horizon memory for human activity understanding and intelligent assistance. | Ann Arbor, MI, USA |
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2021–2022	Carnegie Mellon University <i>Research Intern – Saliency Detection and Virtual Reality</i> Advisor: Prof. David Lindlbauer. Developed RGB-D saliency-detection methods to study how depth information can improve prediction for VR datasets.	United States
2020–2021	Cornell University <i>Research Intern – Pose Reconstruction and Recognition by Wearable Devices</i> Advisors: Prof. Cheng Zhang; Prof. Francois Guimbretiere. Developed wearable machine-learning systems for pose reconstruction, activity classification, facial-expression sensing, and silent-speech recognition.	United States
2020–2021	McGill University <i>Research Assistant – Visual, Audio and Tactile Multimodal Fusion</i> Advisor: Prof. Jeremy R. Cooperstock. Developed automatic multimodal vibrotactile-content creation and contextual audiovisual methods for haptic-audio-visual experiences.	Canada
2019	National University of Singapore <i>Research Intern – Key Point Matching for 3D Non-rigid Shapes</i> Advisor: Prof. Lee Gim Hee. Investigated 3D non-rigid shape key-point matching by learning key-point descriptors.	Singapore
2018	National University of Singapore <i>Research Intern – Unmanned Systems and Real-time 3D Mapping</i> Advisors: Prof. Ben M. Chen; Dr. Shupeng Lai. Developed onboard 3D navigation, GPU-based real-time mapping, and 3D Euclidean signed-distance fields for MAV trajectory generation and planning.	Singapore
2017–2018	Shandong University <i>Research Assistant – Event-Triggered Control for Uncertain 2D Discrete-Time Markov Jump Systems</i> Advisor: Prof. Rongni Yang. Established an event-triggered control scheme for a two-dimensional discrete system using the Roesser model.	China

Professional Service

2026	Local Arrangements Chair , ACM Symposium on User Interface Software and Technology (UIST)
2021–2026	Peer Reviewer , 22 reviews across IMWUT, UbiComp/ISWC, ICMI, CHI, IEEE VR, and VRST
2021	Student Volunteer , ACM MobileHCI
2021	Student Volunteer , IEEE World Haptics Conference (WHC)
2021	Student Volunteer , ACM International Conference on Interactive Media Experiences (IMX)

Teaching

University of Michigan

EECS 571 – Principles of Real-Time Embedded Systems

EECS 598 – Engineering Interactive Systems

EECS 183 – Elementary Programming Concepts

McGill University

COMP 251-001 – Algorithms and Data Structures

COMP 598-001 – Topics in Computer Science: Data Science